Case 02 · Decision log (ADR log): Fire-Sale Cascade Simulator

Companion to the architecture chapter. Each entry follows the ADR (Architecture Decision Record) format: context, decision, alternatives considered and rejected, and consequences. One entry is still open and is explained at the end.

How to read this log

When an architect decides that a wall is load-bearing, they write down why, so that whoever renovates later does not knock it down. An ADR does the same for software: a short record of one decision, what led to it, and what was given up. Records are never deleted. If a decision changes, a new record supersedes the old one, and the old one is marked "superseded".

Statuses: "Accepted" means the system is built according to the decision. "Open" means the question is known and the answer has not been found.

No.	Decision	Status
ADR-001	Computation runs in the browser, not on a server	Accepted
ADR-002	Structure of Arrays instead of an object per agent	Accepted
ADR-003	A persistent pool of workers	Accepted
ADR-004	Publish-only server, no write API	Accepted
ADR-005	No result without a range: every run is also routed to a sweep	Accepted
ADR-006	Ed25519 signatures; the private key never leaves the cloud KMS	Accepted
ADR-007	Reproducible link in the URL fragment; no link for private worlds	Accepted
ADR-008	Calibration runs outside the browser and is published as a signed file	Accepted
ADR-009	Where today's market exposures come from	Open

ADR-001: Computation runs in the browser, not on a server

- **Status:** Accepted
- **Components:** App Shell, Simulation Worker, Local Store

Context

The main users are analysts at regulated institutions. Some will want to load the real balance sheets of entities they supervise. The threat model ranked these as the most sensitive data in the system: a single leak is enough for no regulated body ever to use the tool.

Decision

All computation runs in the user's browser, inside a Web Worker running Pyodide and NumPy. The server only serves static files. There is no channel through which a balance sheet can travel from the browser to the server, and a CSP header enforces this even if code is added by mistake.

Alternatives considered and rejected

- **A compute server:** more powerful, simpler to build, and able to keep history. Rejected because once a balance sheet reaches a server it ends up in a log, a backup or memory, and "we do not keep it" is not an architectural guarantee.
- **A compute server with end-to-end encryption:** rejected because the server would still need to decrypt in order to compute.

Consequences

- **Positive:** there is nothing to steal on the server, and no compute cost on the project side.
- **Negative:** each worker downloads about ten megabytes the first time, and sweeps are limited by browser memory.

ADR-002: Structure of Arrays instead of an object per agent

- **Status:** Accepted
- **Components:** Simulation Worker

Context

Every answer the tool gives is a range, so every click runs hundreds of worlds. The speed of a single run decides whether a sweep takes seconds or minutes.

Decision

Each attribute of all agents lives in one array (Structure of Arrays), and every operation runs on all agents at once as a single NumPy operation. An agent is an index into the arrays, not an object.

Alternatives considered and rejected

- **An object per agent with its own methods:** natural to read. Rejected because the POC measured 1,000 runs in 0.28 seconds with arrays, and objects are tens of times slower. A 512-world sweep would take minutes, and users would settle for single runs.

Consequences

- **Positive:** a full sweep in seconds, even in a browser.
- **Negative:** less natural code. Response rules are written for a single agent and translated into array operations.

ADR-003: A persistent pool of workers

- **Status:** Accepted
- **Components:** Sweep Orchestrator, Simulation Worker

Context

The browser can run several workers in parallel, but each worker loads Python separately, which takes a few seconds and about ten megabytes.

Decision

A pool of workers sized to the number of CPU cores, created once and kept alive for the whole visit. The orchestrator splits each sweep into segments and sends each segment to a different worker (fork-join).

Alternatives considered and rejected

- **A single worker running everything in sequence:** simple. Rejected because a sweep took ninety seconds and users closed the page midway.
- **A new worker per segment:** rejected because loading Python each time cancels the gain from parallelism.

Consequences

- **Positive:** on an 8-core machine a sweep runs about 6 times faster. A crashed worker only costs time, because its segment is re-run by a neighbour and returns the same result (runs are deterministic).
- **Negative:** more browser memory, and a few seconds of waiting the first time.

ADR-004: Publish-only server, no write API

- **Status:** Accepted
- **Components:** Public Library, Library Registry DB, Signing Service

Context

Official worlds must come from the team only, and signed. The question is whether users need to write anything to the server at all.

Decision

The public server serves signed static files only, from a CDN. It has no forms, no write API and no exposed database. Sharing between users happens through links (see ADR-007).

Alternatives considered and rejected

- **A server with an API for saving and sharing worlds:** rejected because a user could upload a fake "official world" that would appear in search next to the signed ones. A write API would also open a channel through which balance sheets could reach the server, contradicting ADR-001.

Consequences

- **Positive:** a tiny attack surface, and everything can be served from a CDN at near-zero cost.
- **Negative:** no sharing between users inside the system.

ADR-005: No result without a range

- **Status:** Accepted
- **Components:** App Shell, Sweep Orchestrator, Sensitivity Analyzer, Visualization

Context

A single run produces one number, and one number looks more certain than it is. The POC showed that between leverage 4 and 5.5 the answer flips from calm to spiral, so the unknown parameters decide the outcome.

Decision

Every single run is automatically routed to a sweep as well, and the screen shows the cascade next to its range. If the sweep is unavailable, the run is labelled "single run, see range", and an image of it cannot be downloaded without the label.

Alternatives considered and rejected

- **Sweeps as an option the user chooses:** rejected because most users would not choose it, and single numbers would end up in slide decks.

Consequences

- **Positive:** every result that leaves the tool carries its uncertainty.
- **Negative:** every click costs hundreds of runs. This is what makes ADR-002 and ADR-003 necessary.

ADR-006: Ed25519 signatures; the private key never leaves the cloud KMS

- **Status:** Accepted
- **Components:** Signing Service, World Loader

Context

An "official" world is a claim: this data reproduces the UK crisis of 2022. Anyone who changes the file in transit can present a fake result in the project's name.

Decision

Every official world and every calibration file is signed with Ed25519 before publication. The private key is held in a KMS and never leaves it. The World Loader checks each signature against the public key; a world whose signature does not match still runs, but is labelled "custom".

Alternatives considered and rejected

- **No signatures, HTTPS only:** rejected because HTTPS protects the transport, not a file forwarded by email.
- **Blocking worlds with a bad signature:** rejected because legitimate users modify worlds to ask "what if". The label is enough, and it appears on every image.

Consequences

- **Positive:** a result cannot be presented as official without a valid signature.
- **Negative:** every library update goes through the signing service, an extra step for the team.

ADR-007: Reproducible link in the URL fragment

- **Status:** Accepted
- **Components:** Link Codec, App Shell

Context

Users want to send a specific run to a colleague. Without a writable server (ADR-004), sharing has to go through a link.

Decision

The link holds the world ID, the seed, the shock and the engine version, in the part after the #. Browsers never send this part to the server. No link is created for a private world, and the interface explains why. A link from another engine version opens with a warning.

Alternatives considered and rejected

- **A link with pre-computed results:** rejected because results in a link are a claim without proof. The link re-runs and gets the same result because runs are deterministic.
- **A link that encodes a private world:** rejected because links are kept in history, chats and email, and the balance sheet would leave the computer.

Consequences

- **Positive:** sharing without a server, without accounts, and without exposing balance sheets.
- **Negative:** a run on a private world cannot be shared. The user can send a screenshot, by their own decision.

ADR-008: Calibration runs outside the browser and is published as a signed file

- **Status:** Accepted
- **Components:** Calibration Cluster, Target Data Store, Signing Service, Public Library

Context

Calibrating against past crises takes tens of thousands of runs. That is too large for a browser, but it needs no user data at all.

Decision

Calibration runs offline on a Dask cluster with pyabc and Numba, using exactly the same engine version. The result, a parameter distribution and a fit report, is signed and published in the library, and official worlds point to it.

Alternatives considered and rejected

- **Calibration in the user's browser:** rejected because it would take hours.
- **Calibration with a separate, faster engine:** rejected because the calibration would then describe a different engine from the one users run.

Consequences

- **Positive:** users get calibrated parameters without waiting, and the cluster never touches user data.
- **Negative:** a cluster to maintain, and a publication process that goes through signing.

ADR-009: Where today's market exposures come from

- **Status:** Open
- **Components:** Simulation Worker, and one box outside the system boundary

Context

To answer the question the tool exists for, what would happen to today's market, the engine needs to know how leveraged each kind of institution is today and who holds what. The POC showed this is the decisive input: between leverage 4 and 5.5 the answer flips from calm to spiral.

Options on the table

- **Buy from a commercial data vendor:** covers only part of the market, is expensive, and usually forbids public use of the data.
- **Ask regulators:** they hold the most accurate data, but releasing it is not in the project's hands, and is usually not allowed.
- **Let the user enter a range:** available immediately, and it is what the system does today. But it is a guess, and the answer comes out conditional: "if leverage is X, this is what happens".

Why it stays open

None of the options is a component the project can build. In the architecture's closure check, this is the only box left outside the lot line, and it is exactly the input the central promise depends on. As long as this record is open, the system can compute crises that already happened, but cannot know what is happening now.